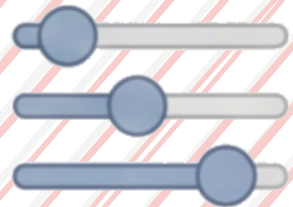
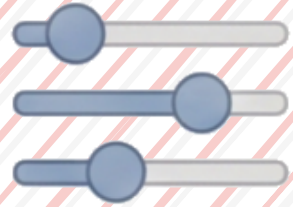


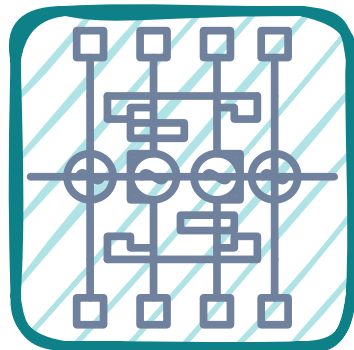
**PARAMETERIZED
LAYOUT
GENERATION**
(Qiskit Metal)



**Geometry
Parameters**



**Topology
Sweeps**



**Generated
Layout
Instance**

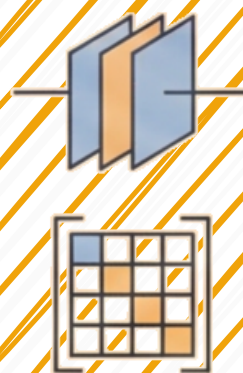
AUTOMATED EM SIMULATION ENGINE

**Eigenmode
Analysis**
(HFSS/pyEPR)



**Raw Modes
& Spectra**
 $(f_i, Q_i, fields)$

**Capacitance
Extraction**
(Q3D/LOM)



**Capacitance
Matrices**
 (C_{in}, C_{eff})

**POST-
PROCESSING
& METRIC
DERIVATION**

$f_r, \kappa/2\pi$

χ, g
 T_1, T_2
estimates

**SQChip-EM
DATASET
REPOSITORY**



**Structured JSON
Annotations**



CSV Index



**Layout Artifacts
(GDS/Rendering)**